Deceptive Humor: A Synthetic Multilingual Benchmark Dataset for Bridging Fabricated Claims with Humorous Content

Disclaimer: This paper includes LLM-generated data on fabricated humor that may unintentionally offend readers.

Anonymous ACL submission

Abstract

This paper presents the Deceptive Humor Dataset (DHD)[1], a novel resource for studying humor derived from fabricated claims and misinformation. In an era of rampant misinformation, understanding how humor intertwines with deception is essential. DHD consists of humor-infused comments generated from false narratives, incorporating fabricated claims and manipulated information using the ChatGPT-4o model. Each instance is labeled with a Satire Level, ranging from 1 for subtle satire to 3 for high-level satire and classified into five distinct Humor Categories: *Dark Humor, Irony, Social Commentary, Wordplay, and Absurdity.* The dataset spans multiple languages including English, Telugu, Hindi, Kannada, Tamil, and their code-mixed variants (Te-En, Hi-En, Ka-En, Ta-En), making it a valuable multilingual benchmark. By introducing DHD, we establish a structured foundation for analyzing humor in deceptive contexts, paving the way for a new research direction that explores how humor not only interacts with misinformation but also influences its perception and spread. We establish strong baselines for the proposed dataset, providing a foundation for future research to benchmark and advance deceptive humor detection models.[2]

1 Introduction

Deceptive humor arises from fabricated or misleading claims, often crafted to provoke laughter or satirical commentary on social, political, or cultural issues. Unlike traditional humor, which primarily relies on exaggeration, absurdity, or irony, deceptive humor distorts false narratives to create a comedic effect. This form of humor thrives in online spaces, where social media facilitates the rapid spread of jokes and satirical content, making it increasingly difficult to distinguish between harmless comedy and harmful misinformation. The challenge in identifying deceptive humor lies in its subjective nature, what appears humorous to one individual might be offensive or misleading to another, especially if the reader is aware of the falsity of the underlying claim. This duality underscores the urgency and complexity of addressing deceptive humor, as its impact varies based on the audience's awareness and interpretation.

In the era of social media, deceptive humor has become an ongoing issue, with a significant portion of online content being generated in response to various claims, both real and fabricated. Given the vast accessibility of social platforms, individuals of all demographics, including children, are exposed to such content. The uncritical consumption and unintentional sharing of deceptive humor can have severe societal repercussions. For instance, a fabricated claim stating, "*Recently, scientists discovered that a steady intake of poison grants superior strength*" may seem humorous at first glance. However, impressionable audiences, such as children or uninformed users, may misinterpret such content, potentially leading to dangerous real-world consequences. This highlights the critical need for robust detection mechanisms to differentiate between deceptive humor and misinformation, ensuring that humor does not inadvertently contribute to spreading harmful narratives.

Given the difficulty in manually collecting real-world samples for deceptive humor, we opted for a structured data generation approach using the ChatGPT-4o model(Hurst et al., 2024)[3]. Identifying deceptive humor in natural discourse is

[1] DHD: Deceptive Humor Dataset

[2] Dataset and other required documents will be publicly available upon acceptance of the paper.

[3] The sample DHD dataset is available for review at the following anonymous GitHub link : https://github.com/NLP-Research07/Deceptive-Humor

inherently complex, requiring knowledge of both the claim and the commenter's intent. Even if a user knows the claim's validity, they may still misinterpret the humor embedded in the comment. This ambiguity makes curating a high-quality dataset from existing social media content challenging. To address this, we systematically generated deceptive humor comments, ensuring control over claim validity and the humorous framing of responses. Moreover, most existing computational models are trained on high-resource languages like English, leaving a significant gap in handling regional languages. With its 1.4 billion people and 22 officially recognized languages[4], India presents a unique opportunity to study multilingual and code-mixed communication. Due to the historical influence of English on India's education system, people often mix their native languages with English, resulting in code-mixed scripts. Our work focuses on building a corpus for deceptive humor detection in English along with four major Indian languages: Telugu, Tamil, Hindi, and Kannada as well as their code-mixed variations. We aim to create a more generalizable and inclusive dataset by including multiple languages. Further details on our data generation methodology are elaborated in later sections of this paper.

Recognizing the need for a dedicated resource to study this phenomenon and to promote the research in this direction, we introduce the *Deceptive Humor Dataset (DHD)*, a novel dataset designed to bridge the research gap in understanding how humor contributes to the propagation of misinformation and the challenges in detecting humorous comments generated from fabricated narratives. While prior research has explored related areas, such as Faux Hate, which refers to hateful comments derived from fake narratives(Biradar et al., 2024), deceptive humor presents a distinct challenge. Unlike explicit hate speech, which users may hesitate to share, deceptive humor often appears harmless and is readily disseminated, making it a more insidious carrier of misinformation. By providing a structured dataset and establishing a new research problem statement, we aim to facilitate further exploration of this critical issue and encourage the development of novel detection methodologies.

[4]https://en.wikipedia.org/wiki/Languages_of_India

Key Contributions of this work:

- **Deceptive Humor: A Novel Intersection of Humor and Misinformation:** This work introduces a novel research problem that fuses humor detection with misinformation, focusing on humor generated from fabricated narratives.
- **First Multilingual and Code-Mixed Synthetic Benchmark for Fact-Aware Humor:** This study presents the first synthetic benchmark corpus designed to analyze fact-aware humor across multiple languages and code-mixed contexts, broadening computational humor research.
- **Evaluating DHD with Diverse PLMs:** We establish a benchmark by evaluating the proposed dataset with widely used open-source models, providing an initial reference point for future research.

The remainder of the article is organized as follows: section 2 reviews existing work on humor and misinformation. section 3 details the dataset construction process and baseline methods for deceptive humor detection. section 5 presents and analyzes the experimental findings. Finally, section 7 summarizes key takeaways and outlines future research directions.

2 Literature Review

Existing research treats humor and misinformation as separate domains, yet their intersection remains largely unexplored. From a psychological and social standpoint, the Interpersonal Humor Deception Model (IHDM) suggests that humor can either reduce self-centered deception and build trust or, if poorly executed, raise suspicions and undermine credibility(Gaspar et al., 2023). Similarly, humor is widely used in advertising to mask deceptive practices, with 73.5% of humorous advertisements containing misleading elements that obscure unethical messaging(Shabbir et al., 2007). These findings highlight humor's dual nature, it can both expose and conceal deception, making it an effective yet ethically complex tool.

From a computational perspective, humor detection in NLP has largely focused on sarcasm (Joshi et al., 2015), irony(Van Hee et al., 2018), and satire(Rubin et al., 2016), but these studies do not address humor generated from fabricated

claims. While humor's role in misinformation detection has been acknowledged(Zhou et al., 2020), prior work treats humor and misinformation as distinct problems, lacking a framework to analyze how humor itself can be a carrier of deceptive content. Additionally, research on fact-checking and misinformation detection(Thorne et al., 2018; Bhardwaj et al., 2020) has primarily focused on textual veracity, without considering how humor can distort factual claims, making detection even more complex. Current humor datasets(Hossain et al., 2019) focus on linguistic features rather than fact-aware humor, limiting their applicability to deceptive humor detection.

3 Methodology

In this section, we describe the process of selecting fabricated claims and generating the Deceptive Humor Dataset (DHD). Using ChatGPT-4o, we create humor-infused comments across multiple languages, ensuring diversity in satire and linguistic variations. We also highlight the role of synthetic data in advancing AI, emphasizing its importance in training robust models and addressing data scarcity in multilingual settings.

3.1 Selection of Fake Claims

The first step in data acquisition involves identifying various topics for data collection. To ensure diversity in the collected data, the authors have selected a range of topics, including entertainment, politics, finance, sports, religion, and health. Following the selection of topics, the next step is to identify fake narratives associated with these topics. These fake narratives are systematically scraped from well-known fact-checking websites such as AltNews[5], Boom FactCheck[6], FactChecker[7], and FACTLY[8]. This approach ensures the reliability and relevance of the data collected for analysis. The selected fake claims are listed in Appendix F.

3.2 Generation of Deceptive Humor corpus

Generating humorous comments based on claims presents significant challenges. Humor often blends seamlessly into conversations, making it difficult to discern without a deep understanding of both the claim and its surrounding context. To address this, we opt to generate data using state-of-the-art (SOTA) generative models. We experimented with several models, including Gemini(Team et al., 2023), Llama(Touvron et al., 2023), Claude, and ChatGPT. While Gemini and Llama perform reasonably well in generating English comments, they struggle with Indic languages, often failing to maintain grammatical structure and contextual coherence. Additionally, Claude does not effectively support the generation of humorous content. Even ChatGPT-4o-mini does not meet our expectations in this regard. Based on a thorough evaluation of the outputs, we identify ChatGPT-4o as the most effective model due to its superior performance, making it the optimal choice for dataset generation.

System Prompt:
You are tasked with generating humorous comments based on fake claims. The comments should be in one of the following languages or code-mixed combinations: Te (Telugu), Hi (Hindi), Ka (Kannada), Ta (Tamil), En (English), Te-En (Telugu-English), Hi-En (Hindi-English), Ka-En (Kannada-English), Ta-En (Tamil-English).

Each generated comment must be labeled with the following:

- **Satire Level:** A value between 1 and 3, where 1 is subtle satire and 3 is high satire.
- **Humor Attribute:** Select one from the following categories:
  - **Irony**
  - **Dark Humor**
  - **Absurdity**
  - **Wordplay**
  - **Social Commentary**

Ensure the comments are complex and human-like, reflecting various humor types and incorporating witty or sharp expressions where appropriate.

Prompt:
Generate a humorous comment based on the following fake claim: `<FAKE_CLAIM>`

The Deceptive Humor Dataset (DHD) is generated using ChatGPT-4o, with carefully structured prompts designed to create humor that feels natural and human-like. To ensure quality and appropri-

[5]https://www.altnews.in/
[6]https://www.boomlive.in/fact-check
[7]https://www.factchecker.in/fact-check
[8]https://factly.in/

ateness, language experts aged above 18, closely monitored the content, making sure it remained engaging without negatively impacting younger audiences. No individuals under 18 are involved in the process, as the dataset is curated with a responsible approach to humor. This careful curation is reflected in the prompt design.

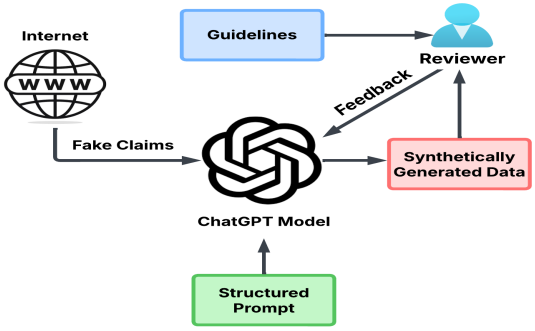

Figure 1: Flowchart for the DHD Data Generation

We enhance data complexity by combining multiple claims and tasking the model to generate comments, producing more nuanced samples. If quality is lacking, we refine it through feedback via instructions or examples to improve human-like text, as shown in the Figure 1.

Role of Synthetic Data in Advancing the AI systems:
The proposed DHD is synthetically generated using the ChatGPT-4o model. A common critique of synthetic data is that pre-trained language models (PLMs) struggle to capture patterns representative of human-generated text. While this argument holds some merit, it is also important to acknowledge that human annotations are influenced by inherent biases shaped by individual mental models(Gautam et al., 2024). The role of synthetic data in AI research has become increasingly significant in recent years. Numerous top institutions, including Hugging Face[9], and various companies are actively researching the use of synthetic data and have recently released synthetic data generators[10] to support this effort. Notably, the Phi-4 model[11] a state-of-the-art(SOTA) open model has effectively utilized synthetic data as a core component of its training regimen.

In light of these advancements, it is evident that synthetic data, once a point of contention, has emerged as an invaluable tool in the evolution of modern AI systems. Its integration plays a pivotal role in improving model generalization, enhancing capabilities, and addressing data limitations. Furthermore, synthetic data offers the added advantage of mitigating biases that may be introduced by human annotators, thereby fostering the development of more robust, unbiased, and versatile AI models. This transformative approach not only expands the scope of AI applications but also ensures greater fairness and reliability in model outcomes.

3.3 Dataset Description

The proposed DHD consists of 9,000 synthetically generated humorous comments, carefully curated to ensure linguistic diversity and humor variation. The dataset is split into 7,200 comments for training, 900 for validation, and 900 for testing as shown in Table 1. Each comment is labelled with a satire level ranging from 1 to 3, where 1 represents subtle satire and 3 denotes highly exaggerated satire. Additionally, every comment is assigned a humor attribute from one of the five predefined categories: Irony, Absurdity, Social Commentary, Dark Humor, and Wordplay.

Statistic	Train	Validation	Test
Total Samples	7,200	900	900
Satire Level Distribution			
Low Satire	2,092	251	253
Moderate Satire	3,125	378	388
High Satire	1,983	271	259
Humor Attribute Distribution			
Irony	2,180	270	284
Absurdity	1,639	196	208
Social Commentary	1,217	149	142
Dark Humor	1,101	154	129
Wordplay	1,063	131	137

Table 1: DHD Distribution

A key aspect of the DHD is its linguistic diversity. Along with English, it includes comments in four major Indic languages: Telugu, Hindi, Kannada, and Tamil, along with their code-mixed versions (Te-En, Hi-En, Ka-En, Ta-En). This ensures a rich and varied dataset that captures the nuances of humor across multiple languages and cultural contexts. The structured labeling enables a comprehensive analysis of humor in NLP systems, fostering advancements in computational humor understanding, particularly in multilingual and

[9] https://huggingface.co/
[10] https://huggingface.co/blog/synthetic-data-generator
[11] https://huggingface.co/microsoft/phi-4

code-mixed settings. A detailed description of the dataset is presented in Table 2.

Description of the Labels:

Satire Level: This label quantifies the degree of satire in the generated comment.

- **Low Satire (1):** The humor is subtle and lightly satirical, often resembling real-world statements with a mild twist.
- **Moderate Satire (2):** The humor is more evident, incorporating exaggeration and sarcasm while maintaining a balance between reality and absurdity.
- **High Satire (3):** The humor is strongly exaggerated and overtly satirical, often making use of extreme irony or absurd distortions of reality.

Humor Attribute: This label categorizes the type of humor used in the comment.

- **Irony:** A form of humor where the intended meaning contrasts sharply with the literal meaning, often exposing contradictions or unexpected outcomes.
- **Absurdity:** Humor that thrives on exaggeration, illogical scenarios, or unrealistic premises to create an amusing effect.
- **Social Commentary:** Humor that critiques, mocks, or highlights societal or cultural issues, often with a satirical or thought-provoking angle.
- **Dark Humor:** Humor that deals with morbid, taboo, or controversial topics in a way that might be unsettling but still amusing.
- **Wordplay:** Humor that relies on clever linguistic constructs, including puns, double meanings, and phonetic playfulness.

4 Experimental Setup

In this section, we discuss the experimental setup used to evaluate the proposed DHD. The dataset is tested across different model architectures, including Encoder-Only Models, Encoder-Decoder Models, and Large Language Models(Brown et al., 2020) with Zero-Shot, Few-Shot, and QLoRA(Dettmers et al., 2024) fine-tuning.

Encoder-Only models:

To establish strong baseline results, we evaluate on several transformer-based models(Vaswani et al., 2017). **BERT**(Devlin et al., 2019) is a bidirectional transformer trained using masked language modeling and next sentence prediction. **DistilBERT**(Sanh et al., 2019) is a distilled version of BERT that is smaller, faster, and lighter, achieving similar performance while being more efficient. **mBERT** extends BERT to multilingual settings, enabling cross-lingual tasks. **XLM-RoBERTa**(Conneau et al., 2019) is a large multilingual language model trained on 100 languages, enhancing cross-lingual understanding. **DeBERTa**(He et al., 2020) improves upon BERT and RoBERTa by using disentangled attention and enhanced mask decoder mechanisms. **ALBERT**(Lan et al., 2019) is a lightweight model that reduces parameters through cross-layer parameter sharing and factorized embedding parameterization. Lastly, **XLNet**(Yang et al., 2019) leverages permutation-based training to address the limitations of BERT, capturing bidirectional context while maintaining autoregressive properties.

Encoder-Decoder models:

Following the evaluation of encoder-only models, we also assess several transformer-based encoder-decoder models. **BART**(Lewis et al., 2020) is a denoising autoencoder that combines a bidirectional encoder with an autoregressive decoder, excelling in text generation and comprehension tasks. **mBART**(Liu et al., 2020) extends BART to a multilingual context, pre-trained as a sequence-to-sequence denoising autoencoder on large-scale monolingual corpora across multiple languages, making it effective for neural machine translation. **T5**(Raffel et al., 2020) frames all NLP tasks as text-to-text transformations, enabling a unified approach to various tasks such as translation, summarization, and question answering.

Large Language Models:

After computing the results with Encoder-Only and Encoder-Decoder Models, we evaluated several open-source large language models on the proposed DHD. Google Gemini(Team et al., 2024) is known for its enhanced reasoning abilities, offering high efficiency in processing complex tasks. Meta Llama(Touvron et al., 2023), provides open-source LLMs optimized for diverse NLP applications, with a focus on scalability and efficient inference across different hardware platforms. Microsoft Phi-4(Abdin et al., 2024) incorporates synthetic data in its training set, which

Language	Total Records	Satire Level			Humor Attribute				
		1	2	3	Absurdity	Dark Humor	Irony	Social Commentary	Wordplay
Train Data									
English	877	190	323	364	243	147	297	101	89
Telugu	819	300	345	174	178	111	321	162	47
Hindi	822	264	386	172	166	90	285	169	112
Kannada	771	268	348	155	203	95	218	185	70
Tamil	725	231	343	151	183	99	238	143	62
Telugu-English	815	194	345	276	195	153	206	114	147
Hindi-English	831	219	349	263	122	137	233	130	209
Kannada-English	768	199	329	240	189	162	175	110	132
Tamil-English	772	227	357	188	160	107	207	103	195
Validation Data									
English	112	21	31	60	38	23	31	12	8
Telugu	110	33	49	28	19	19	48	23	1
Hindi	100	38	40	22	20	9	30	24	17
Kannada	84	30	30	24	18	10	27	21	8
Tamil	88	29	43	16	17	11	34	16	10
Telugu-English	104	23	49	32	20	23	32	10	19
Hindi-English	99	23	53	23	11	12	24	21	31
Kannada-English	108	27	45	36	32	25	17	14	20
Tamil-English	95	27	38	30	21	22	27	8	17
Test Data									
English	110	21	37	52	32	20	39	11	8
Telugu	117	38	51	28	33	15	43	18	8
Hindi	80	23	47	10	22	6	27	16	9
Kannada	81	29	32	20	22	15	19	17	8
Tamil	100	32	49	19	28	9	44	13	6
Telugu-English	114	26	44	44	23	20	25	15	31
Hindi-English	95	25	41	29	14	13	23	17	28
Kannada-English	115	32	55	28	21	19	36	16	23
Tamil-English	88	27	32	29	13	12	28	19	16

Table 2: Distribution of Satire Levels and Humor Attributes Across Languages (Appendix E provides a visualization of these distributions).

aids in improving its performance, particularly in mathematical reasoning tasks, by providing more diverse and challenging training scenarios.

We evaluated LLMs on the DHD using *Zero-Shot* and *Few-Shot* approaches. Zero-Shot required models to make predictions without task-specific examples, relying on pre-trained knowledge, while Few-Shot provided limited labeled examples for better contextual adaptation. To enhance performance, we fine-tuned the LLMs using QLoRA (Dettmers et al., 2024), which applies low-rank adaptations efficiently. This fine-tuning helped models incorporate domain-specific knowledge and improve task-specific performance. Details on prompt generation, inputs, and the experimental setup for Zero-Shot, Few-Shot, and QLoRA fine-tuning are in Appendix B and Appendix D.

5 Results and Discussion

In the results for Satire Level Classification, the mBART (Liu et al., 2020) outperformed other models, while for the Humor attribute, the BERT (Devlin et al., 2019) performed better. In Zero-Shot and Few-Shot settings, LLMs exhibited bias by disproportionately favoring certain labels while underutilizing others, with some labels receiving little to no representation in predictions, leading to significantly lower scores, as shown in Table 5. To enhance domain knowledge and improve performance, we employed the QLoRA approach(Dettmers et al., 2024), which led to better results than the Zero-Shot and Few-Shot versions. However, even with QLoRA's enhancements, language models still struggle with deceptive humor a highly nuanced category requiring contextual and cultural understanding. The subtle and ambiguous nature of deceptive humor makes it difficult to distinguish from other humor types or even non-humorous text, limiting generalization. This highlights the need for further research to develop more robust techniques, given its broader implications in misinformation detection, social media content moderation, and computational humor studies. The detailed error analysis of some

challenging samples is discussed in Appendix H, providing further insights into the limitations and failure cases of existing models.

Our findings reveal a fundamental challenge: existing models, which perform well on humor detection and misinformation classification individually, struggle significantly when dealing with deceptive humor. This is because deceptive humor is not merely about understanding humor; it also requires fact verification and intent recognition, which traditional humor detection models lack. While humor classification has been explored using datasets such as SemEval-2017 Task 6 on Humor Detection (Potash et al., 2017), Humicroedit (Hossain et al., 2019), these datasets primarily focus on linguistic humor rather than humor derived from fabricated claims. Similarly, misinformation detection datasets such as FEVER (Thorne et al., 2018), LIAR (Wang et al., 2017), and Hostile Dataset(Bhardwaj et al., 2020) focus on textual veracity but do not account for the nuances of humor distorting false narratives. Although synthetic humor datasets have been introduced, such as the Unfun dataset (Horvitz et al., 2024), which edits humorous texts to make them non-humorous for improved humor detection, their primary focus is on humor manipulation rather than humor intertwined with deception. This highlights the increased complexity of deceptive humor, which combines linguistic ambiguity, misinformation, and humor-specific reasoning, making it a more challenging task than standard humor detection or fact-checking alone.[12]

Unlike prior work that treats humor and misinformation as separate problems, our findings highlight how humor itself can actively propagate misinformation, further complicating detection. This introduces a fundamentally new challenge that requires both linguistic and factual reasoning, where existing models struggle. Our results in Table 5 confirm this gap even fine-tuned transformer models and LLMs fail to generalize effectively across different humor attributes and satire levels in the proposed DHD.These limitations emphasize the urgent need for novel approaches that integrate both humor understanding and misinformation detection to effectively handle deceptive humor.

[12]ColBERT Humor Detection Dataset: https://huggingface.co/datasets/CreativeLang/ColBERT_Humor_Detection

Results Reproducibility:
The hyperparameters used in our experiments are presented in Tables Table 3 and Table 4, specifying the configurations for different settings. These details ensure the reproducibility of our results by providing a clear reference for the training and evaluation setups.

Hyperparameter	Value
Max Len	128
Batch Size	16
Epochs	5
Learning Rate	3e-5
Drop Out	0.3

Table 3: Hyperparameter settings for Encoder-Only and Encoder-Decoder Models.

Hyperparameter	Value
LoRA Rank (r)	16
LoRA Alpha	64
LoRA Dropout	0.2
Target Modules	{k_proj, q_proj, v_proj}
Learning Rate	2e-5
Batch Size	8
Epochs	3
Weight Decay	0.01

Table 4: Hyperparameter settings for the LLM Models.

6 Human Evaluation

To evaluate the quality of the synthetically generated dataset, we conducted a human evaluation with native speakers and language experts familiar with the fake claims to validate the generated records. Given the nature of our work, we selected evaluators over the age of 18 from diverse regions, as shown in Table 6.

Evaluators received clear guidelines and all necessary materials before the evaluation began. To ensure their understanding of the task, we first conducted a mock evaluation with a small sample set. Based on their performance, we selected the most suitable evaluators. They rated the labels assigned by the ChatGPT-4o model on a scale from 0 to 5, where 0 indicated the poorest match and 5 the best. To ensure diverse label representation, we randomly selected 50 records per language, totaling 450 manually validated samples across all languages. As a token of appreciation, evaluators

Model	Satire Level				Humor Attribute				Parameters
	Accuracy	F1-Score	Precision	Recall	Accuracy	F1-Score	Precision	Recall	
Encoder-Only									
BERT(Devlin et al., 2019)	49.44	46.47	49.22	46.19	**40.44**	34.73	**44.37**	34.42	110M
DistilBERT(Sanh et al., 2019)	49.44	45.60	49.42	45.60	37.89	34.58	38.95	34.09	66M
mBERT	50.44	50.00	49.98	**50.06**	36.00	**35.77**	37.19	35.00	110M
XLM-RoBERTa(Conneau et al., 2019)	49.33	48.87	48.86	48.93	34.44	33.70	33.95	33.60	125M
DeBERTa(He et al., 2020)	47.89	42.52	49.07	43.29	37.33	28.76	30.00	31.51	184M
ALBERT(Lan et al., 2019)	46.33	43.52	46.24	43.25	38.89	33.33	39.91	33.56	11M
XLNet(Yang et al., 2019)	50.44	46.72	50.84	46.63	36.78	25.49	31.41	28.52	117M
Encoder-Decoder									
BART(Lewis et al., 2020)	49.11	46.44	49.34	46.09	37.89	33.84	37.93	33.93	407M
mBART(Liu et al., 2020)	**51.00**	**50.24**	**51.04**	49.82	36.11	35.60	36.32	**35.22**	680M
T5(Raffel et al., 2020)	46.78	43.64	46.37	43.53	37.67	28.48	36.02	30.50	738M
Decoder-Only (Zero-Shot)									
Gemma(Team et al., 2024)	35.78	28.76	30.03	31.46	23.44	14.66	13.44	19.47	7000M
Llama-3.1(Touvron et al., 2023)	27.78	19.23	27.10	32.73	21.11	16.65	20.55	20.60	8000M
Phi-4(Abdin et al., 2024)	42.40	20.35	20.93	32.90	15.00	11.35	10.33	20.22	14000M
Decoder-Only (Few-Shot)									
Gemma	43.11	20.08	14.37	33.33	31.56	9.59	6.31	20.00	7000M
Llama-3.1	28.78	14.90	9.59	33.33	29.67	12.08	10.04	19.93	8000M
Phi-4	28.78	14.90	9.59	33.33	14.33	5.01	2.87	20.00	14000M
Decoder-Only (QLoRA Fine-Tuned)									
Gemma	34.00	27.00	40.00	33.00	24.00	21.00	26.00	24.00	7000M
Llama-3.1	30.00	24.00	34.00	35.00	22.00	18.00	21.00	21.00	8000M
Phi-4	35.00	29.00	34.00	34.00	26.00	12.00	35.00	18.00	14000M

Table 5: Baseline Metrics of Models Across Satire Levels and Humor Attributes. The top results are represented in **bold**, and the second-best results are underlined.

were granted free access to GPU resources. This evaluation highlights the alignment between model-generated labels and human judgment, providing valuable insights into dataset quality. The results are summarized in Table 7.

Language	Education	Gender	State
English	B.Tech	Male	Andhra Pradesh
Telugu*	B.Tech	Male	Andhra Pradesh
Hindi*	PhD Candidate	Male	Delhi
Kannada*	PhD Candidate	Male	Karnataka
Tamil*	B.Tech	Male	Tamil Nadu
Country: India			

Table 6: Profile of Human Evaluators Across Languages (* indicates Indic languages along with their code-mixed variants).

Language	Avg Satire Score	Avg Humor Score	Overall Avg. Score
English	4.21	4.14	4.17
Telugu	3.67	4.00	3.83
Hindi	3.50	3.60	3.55
Kannada	3.00	2.80	2.95
Tamil	3.00	3.87	3.43
Telugu-Eng	4.20	4.00	4.10
Hindi-Eng	4.00	4.00	4.00
Kannada-Eng	3.00	3.00	3.00
Tamil-Eng	3.74	4.34	4.04

Table 7: Human Evaluation results for Satire, Humor, and Average scores across various language variants.

The human evaluation results reveal that English comments receive higher ratings, likely due to the ChatGPT-4o model's extensive training on English compared to languages such as Telugu, Hindi, Tamil, and Kannada, particularly in their native scripts. Notably, the code-mixed versions of these languages achieve relatively higher scores than their native-script counterparts. This observation suggests that the model demonstrates greater proficiency in generating Indic languages in a code-mixed format, reflecting an imbalance in training data coverage and script-specific linguistic capabilities.

7 Conclusion and Future work

In this study, we proposed a novel research direction at the intersection of misinformation and humorous content, emphasizing how humor can act as a major vehicle for spreading misinformation. This approach highlighted the need to better understand the interplay between humor and misinformation, which is often overlooked in existing research. The study underscored the importance of addressing this issue as it plays a significant role in shaping public perception and influencing societal narratives. In future work, there is a pressing need to focus on detecting and mitigating deceptive humor, which has become increasingly prevalent. Leveraging advanced methods in nlp will be essential for tackling this challenge, given its growing impact on misinformation in contemporary times.

8 Limitations

The proposed work relies on synthetically generated data, which may not fully replicate the complexity and nuance of human-generated text. While synthetic data allows for controlled and scalable dataset creation, it often lacks the unpredictability and cultural context inherent in naturally occurring humor. Additionally, generative models may introduce biases or produce humor that does not always align with human expectations. However, we mitigate these concerns by carefully designing prompts and examples to approximate human-like complexity, ensuring that the dataset remains linguistically diverse and contextually relevant.

Despite these limitations, the use of synthetic data has gained significant traction in NLP research, demonstrating its ability to mimic human-generated text effectively while addressing data scarcity issues. Studies have shown that synthetic data can be comparable to human-labeled data in various NLP tasks, particularly in low-resource language settings(He et al., 2022). Large-scale models have successfully leveraged synthetic datasets to improve generalization and task-specific performance(Veselovsky et al., 2023). Furthermore, recent work has demonstrated that synthetic data can enhance model robustness by introducing controlled variations that mirror real-world complexities(Liu et al., 2024). Given the increasing reliance on LLM-generated datasets across NLP domains, our approach aligns with ongoing efforts to leverage synthetic data.

9 Ethical Considerations

The Deceptive Humor Dataset (DHD) is released for academic and research purposes only. It intends to facilitate studies at the intersection of misinformation and humor. Any commercial use, application in real-world misinformation campaigns, or use in generating deceptive content for malicious purposes is strictly prohibited. Researchers utilizing this dataset should ensure ethical compliance and consider potential biases inherent in AI-generated content.

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A Technical Appendix

B Zero Shot Prompts

Zero Shot prompting is used to classify a comment based on its humor or satire level, without requiring any labeled examples in the prompt. This approach allows the model to generalize based on the provided categories.

B.1 Zero Shot Prompt for Satire Level

The following prompt is used to classify the level of satire in a comment. The categories are defined as follows:

```
PROMPT = "Classify the following comment based on the Satire Level. The categories
are as follows:
1 refers to Low Satire,
2 refers to Medium Satire,
3 refers to High Satire."
```

B.2 Zero Shot Prompt for Humor Attribute

The following prompt is used to classify the humor type of a comment. The categories are defined as:

```
PROMPT = "Classify the following comment based on the humor attributes. The
categories are as follows:
0 refers to 'Irony',
1 refers to 'Absurdity',
2 refers to 'Social Commentary',
3 refers to 'Dark Humor' and
4 refers to 'Wordplay'"
```

C Few Shot Prompts

Few Shot prompting is used when we provide a few labeled examples along with the query to help the model understand the context better. It typically results in more accurate predictions, as the model has specific examples to learn from.

C.1 Few Shot Prompt for Satire Level

The following prompt is for classifying the satire level in comments with the inclusion of example data to provide better insights:

```
PROMPT = "Classify the following comment based on satire levels:
1: Low Satire
2: Medium Satire
3: High Satire
For better insights, consider the following examples:
Comment: If Pathan is spreading anti-Hindu sentiment, then I must be spreading
good vibes by doing absolutely nothing.
Given Satire Level: 1
Comment: Adipurush's CGI is the only thing that unites both atheists and believers
everyone hates it equally.
Given Satire Level: 2
Comment: If spitting on food spreads COVID, chefs should get a hazard pay upgrade!
Given Satire Level: 3 "
```

C.2 Few Shot Prompt for Humor Attribute

The following prompt is for classifying the humor attribute in comments, including a few examples to improve the model's understanding:

```
PROMPT = "Classify the following comment based on humor attributes:
0: 'Irony',
1: 'Absurdity',
2: 'Social Commentary',
3: 'Dark Humor',
4: 'Wordplay'.
For better insights, consider the following examples:
Comment: If spitting on food spreads COVID, chefs should get a hazard pay upgrade!
Given Humor Attribute: Dark Humor
Comment: Is China spreading coronavirus as a biological weapon or just making
2020 more interesting?
Given Humor Attribute: Absurdity
Comment: If coronavirus is China's biological weapon, someone should really review
the script!
Given Humor Attribute: Irony
Comment: If Pathan is spreading anti-Hindu sentiment, then I must be spreading
positive energy in my sleep.
Given Humor Attribute: Wordplay
Comment: China bio weapon lo chesindi, mana andarni memers ga marchindi!
Given Humor Attribute: Social Commentary "
```

D QLoRA Approach: Prompt Generation Functions

QLoRA is a fine-tuning approach for large language models that uses low bit precision, maintaining performance while reducing computational costs. The use of structured prompts in QLoRA helps the model better understand and classify text, improving its ability to handle tasks like humor and satire classification. Below are the key functions and label mappings.

D.1 Prompt Generation Functions

Training Data Prompt

```
generate_train_prompt(text, label):
def generate_train_prompt(text, label):
      return (f"Comment: {text} | The (Satire Level)/(Humor Attribute) assigned
to this comment is {label}, which corresponds to the humor type:
{label_mapping[label]}.")
```

Validation/Test Data Prompt

```
generate_eval_prompt(text):
def generate_eval_prompt(text):
      return f"Analyze the following comment: {text}";
```

D.2 Label Mappings

Label Mapping for Satire Level

```
label_mapping = {
1: "Low Satire",
2: "Medium Satire",
3: "High Satire",
}
```

Label Mapping for Humor Attribute

```
label_mapping = {
0: "Irony",
1: "Absurdity",
2: "Social Commentary",
3: "Dark Humor",
4: "Wordplay"
}
```

E Heatmap Analysis of Satire and Humor Across Languages

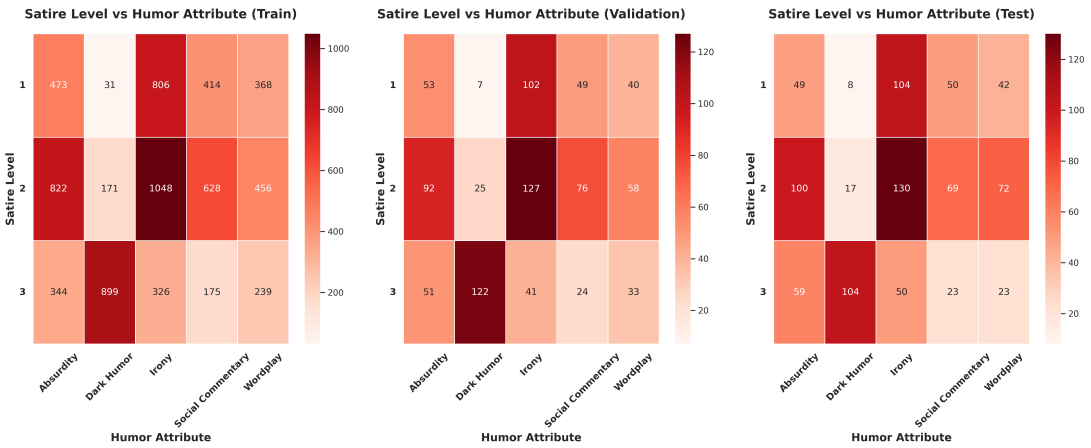

Figure 2: Satire Level vs Humor Attribute. The color intensity represents the frequency distribution across datasets.

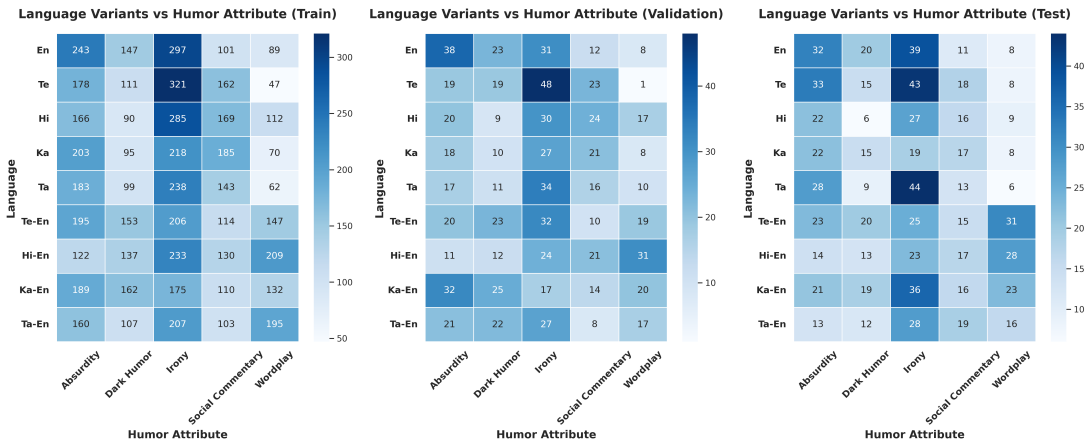

Figure 3: Language Variants vs Humor Attribute. The heatmap highlights humor variations across languages.

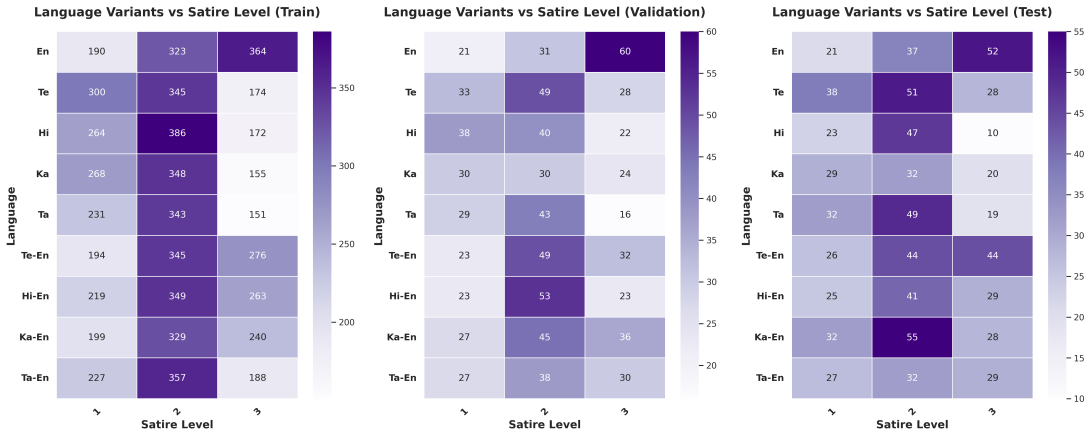

Figure 4: Language Variants vs Satire Level. The color intensity represents frequency distribution, showing how different languages align with satire levels.

F Claims Supporting this Research Work

Claims Supporting this Research Work:

- World Health Organization (WHO) declares eating meat causes cancer.
- India has replaced the Indian Rupee with Digital Currency Completely.
- Muslims are spitting on food to spread COVID-19.
- Hindu temples in India are being destroyed and replaced with mosques.
- African immigrants are responsible for rising crime in Europe.
- The LGBTQ+ community is pushing for legalizing pedophilia.
- Specific vaccines cause sterility in minority communities.
- The COVID-19 vaccine turns you into a zombie after two years.
- China is actively exporting 'fake rice' made from plastic to African countries.
- Salaar had a crossover with KGF, both are part of the cinematic universe.
- Bollywood is spreading anti-hindu sentiments through its movies like Pk, Adipurush and pathan.
- The Earth Is Flat.
- Sharks Are Immune to Cancer.
- Pyramids Were Built by Aliens.
- NASA confirms 15 days of darkness due to planetary alignment.
- The Taj Mahal was originally a Hindu temple named Tejo Mahalaya.
- Vaccines contain microchips for surveillance.
- Muslims in Kerala are forcing non-Muslims to convert to Islam with the help of the government.
- Chinese companies are secretly building surveillance networks in Indian cities.
- The Indian government is introducing a 'one-child policy' to control the population.
- Muslim community in India is receiving special treatment in government welfare schemes.
- Bill Gates created the COVID-19 virus to implant tracking devices via vaccines.
- India's government is secretly constructing detention camps for Muslims in Assam.
- Justice Muralidhar was transferred overnight by the central government due to his critical remarks against the Delhi Police during the Delhi riots.
- Muslim youth beaten up, detained for urinating on Kumbh banner.
- 300-year-old St. Joseph's Church in Sugnu, Imphal. 'BJP-supporting extremists' had set it on fire.
- Kukis bombing Meitei areas in Manipur.
- China spreading coronavirus as a biological weapon.
- Tablighis spreading coronavirus as jihadi propaganda.
- Migrant workers beaten during the coronavirus lockdown.
- Farmer protest is sponsored by Khalistan.
- GPS chip installed in 2000 rupee notes.
- Wrestler Sakshi Malik being pinned down by cops.
- AAP offers monetary relief only to Muslim victims of Delhi riots.
- Shaheen Bagh women getting paid.
- Posters of Hindu deities burnt during CAA protests.
- COVID-19 booster shots lead to AIDS.
- All vaccinated people will die within 2 years.
- Tablighi Jamaat members roaming naked in an isolation ward.
- Muslim man threw an Indane gas cylinder on the tracks in Haldwani.
- Station master at Bahanaga Bazar station is a Muslim and fled the accident site.
- Hindu temple near Bahanaga Bazar station portrayed as a mosque, falsely claiming the Muslim community is behind the train accident.
- RSS training Hindus to entrap Muslim girls.
- Claims of Dhoni and Kohli donating to the Odisha train accident circulate on social media.
- Pathan is spreading anti-Hindu sentiment.

- Adani Group is accused of fraud and stock manipulation by Hindenburg Research.
- Prabhas dating Anushka
- Banner was seen in front of the gate of Tihar Jail reading “Kejriwal will come again”.
- Adani publicly admitted to paying bribes, suggesting the practice extended to India.
- Stalin’s photo displayed at Times Square in New York.

G Sample Records of Deceptive Humor

Language	Comment	Satire Level	Humor Attribute
En	Next up: redefining gravity because someone feels it’s oppressive.	3	Absurdity
En	Mosques near stations now double as train wreck centers? Innovative blaming!	3	Dark Humor
En	The Earth is so flat that NASA has been fooling us with Photoshop for decades.	2	Irony
En	Even the beggars now ask, ’Do you have Paytm, sir? No cash, please!’	1	Social Commentary
En	They say sticks and stones break bones, but in lockdown, it was just ’discipline’!	2	Wordplay
Te-En	So now Khalistan is getting into the protest business. Can I book a protest for my birthday?	2	Absurdity
Te-En	Temples ni mosques chesthunnaru anta, next Ramayan ni alochinchi Game of Thrones laga chesthara?	3	Dark Humor
Te-En	Corona ni jihadi tool ga use chesthunaru anta, inkem cheptaru? Wi-Fi lo use chesi live cheyyadam?	2	Irony
Te-En	Arey mamsam tintaru ani cheppi kotha rule pettara WHO?	1	Social Commentary
Te-En	Tablighi Jamaat spreading COVID-19 like it’s a secret mission, script lo plot twist thappa!	3	Wordplay
Hi-En	Bhai, anthem ke jagah Azan bajegi kya ab?	2	Absurdity
Hi-En	Bhai, China ke rice ke saath ek ’Recycle or Die’ sticker bhi milega!	3	Dark Humor
Hi-En	China ka corona bio weapon? Aur hum lockdown comedy mein atak gaye!	3	Irony
Hi-En	Arre bhai, ab roadside vendors bhi bolenge ’QR lagao ya aage badho’!	2	Social Commentary
Hi-En	Plastic rice mein agar fiber hai, toh diet ke liye perfect hoga!	3	Wordplay
Ka-En	Dhoni Instagram alli photo haakidre issue complete solve agathe anta!	2	Absurdity
Ka-En	Bill Gates created COVID-19 to implant tracking devices, well, where’s the tracking device for ‘common sense’?	3	Dark Humor
Ka-En	Digital rupee anta, but bank queue sullu badlu illa yenadre?	1	Irony
Ka-En	Gas cylinder throw madi Haldwani summit-anna join agidru anta!	1	Social Commentary
Ka-En	Flat earth andre nimma GPS ge flat map kodutha?	1	Wordplay
Ta-En	Bro unga ooru marriage la gift QR code la thaane kudupaanga?	2	Absurdity
Ta-En	World-a kaapatharadhukku virus-a anupa China, evvalavo innovation!	3	Dark Humor
Ta-En	Church-ai set pannina history save aaguma?	1	Irony
Ta-En	Temple oru ’creative hub’ nu solra trend WhatsApp-la!	1	Social Commentary
Ta-En	Vaccines-ku microchip install panna? Mari app store-la update varuma?	1	Wordplay

Table 8: Deceptive humorous comments across different languages with their corresponding satire levels and humor attributes (additional data can be accessed through the given anonymous GitHub Link.).

H Error Analysis of Model Misclassifications

We analyzed model misclassifications in satire level and humor attributes. The key patterns observed are:

- **Irony misclassified as Social Commentary**
  Example: "China is exporting plastic rice—save the environment, eat plastic!"
  The model interpreted the contradiction as a critique rather than irony.
- **Exaggeration mistaken for Dark Humor**
  Example: "WHO bans meat eating forever to reduce climate change."
  Misclassified due to the serious undertone, despite being wordplay.
- **Absurdity confused with Irony**
  Example: "Flat Earth believers demand GPS satellites stop lying to them!"
  The model failed to distinguish absurdity from sarcasm.
- **Subtle satire labeled as High Satire**
  Example: "Even beggars now ask, ’Do you have Paytm, sir?’"
  Mild satire misinterpreted as extreme.
- **Social Commentary misclassified as Irony**
  Example: "Farmers’ protests are a birthday package deal—pay extra for a celebrity endorsement!"
  The model overlooked the political critique and labeled it as sarcasm.

These errors highlight the difficulty in differentiating between satire intensities and humor attributes, indicating the need for improved contextual understanding and fact-aware to understand and detect the deceptive humor comments.